

Digital Forensics & Incident Response

UNIT 3 - Windows & Linux Forensics

Jigar Gajjar
Vice President – Academia
TechD Cybersecurity Limited

Contents

- **Windows File System Forensics**
- **NTFS and Master File Table**
- **Windows Registry Forensics**
- **Windows Event Log Analysis**
- **Windows Operating System Artefacts**
- **Linux File System Forensics**
- **Linux System and Authentication Logs**
- **Linux User and Command-Line Artefacts**
- **Linux Scheduled Tasks and Network Artefacts**
- **Correlation and Timeline Analysis**



Windows File System Forensics



Introduction to Windows & Linux Forensics

Windows and Linux forensics involves the systematic examination of operating-system data and artefacts to reconstruct user activity, system activity, authentication events, application usage and other events relevant to an investigation.

Operating systems continuously generate information through file systems, registries, logs, user activity and system processes. These artefacts can provide valuable information about **what happened, when it happened and how the system was used**.

Major Areas

- Windows File System Forensics
- Windows Registry Forensics
- Windows Event Log Analysis
- Windows Operating-System Artefacts
- Linux Forensics



Windows File System Forensics

NTFS (New Technology File System) is the primary file system used by modern Windows operating systems.

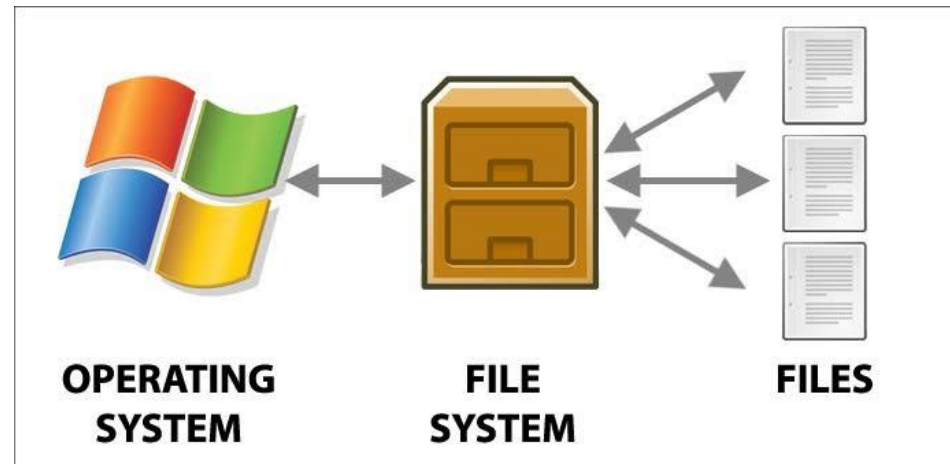
Forensic examination of NTFS focuses on understanding how files and directories are stored and how file-system metadata can provide information about user and system activity.

Important NTFS structures include:

Master File Table (MFT)

MFT records

- File and directory metadata
- File timestamps
- Alternate Data Streams
- Deleted-file information



Forensic Importance

NTFS metadata can help investigators establish the existence, location, modification and other characteristics of files.



Master File Table (MFT)

The **Master File Table (MFT)** is a central structure in NTFS that maintains records associated with files and directories on the volume.

Each file or directory is represented by an **MFT record** containing metadata and information used by NTFS to manage that object.

Forensic Information

- MFT records can provide information such as:
- File name
- File location
- File size
- File attributes
- Timestamps
- File allocation information

Importance

MFT analysis can help investigators understand **file-system activity and the history of files**, including information associated with deleted files.



MACB Timestamps

MACB timestamps represent four important categories of file-system time information.

M - Modified

Indicates when the contents of a file were modified.

A - Accessed

Indicates when the file was accessed or read.

C - Changed

Represents a change to file-system metadata or attributes.

B - Born / Created

Represents when the file was created.

Forensic Importance

Timestamps help investigators establish the **sequence and timing of events** and correlate file activity with other evidence. However, timestamps should be interpreted carefully because system operations, configuration and other factors can affect them.



Alternate Data Streams

Alternate Data Streams (ADS) are additional streams of data associated with a file in the NTFS file system. A file can contain a primary data stream as well as additional named streams.

Forensic Importance

ADS can be used for legitimate purposes, but they can also be abused to **store information that may not be immediately visible during ordinary file browsing.**

Therefore, forensic examination should consider the possibility of data existing outside the primary file content.

Investigation Focus

- Identify files containing alternate streams
- Examine the stream names
- Determine the type of data stored
- Correlate the data with other evidence



File Deletion in NTFS

Deleting a file does not necessarily mean that its underlying data is immediately destroyed.

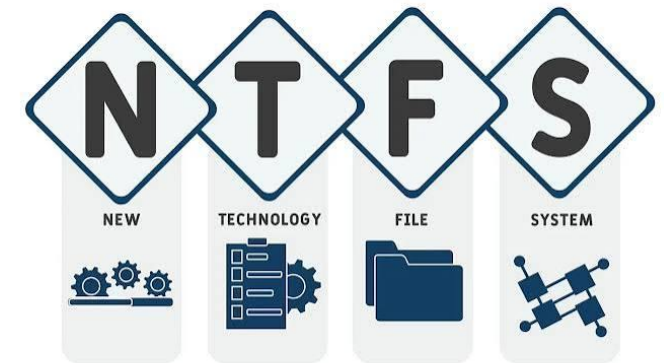
When a file is deleted, its MFT entry can be marked as **inactive**, while the storage space associated with the file becomes available for future use.

The actual data may remain recoverable until the relevant storage clusters are reused or overwritten.

Simplified Process

File exists → File is deleted → MFT entry becomes inactive → Space becomes available → Data may remain → Clusters may eventually be reused → Forensic Importance

This behaviour makes deleted-file recovery possible under suitable conditions.



Windows Registry Forensics



Windows Registry Forensics

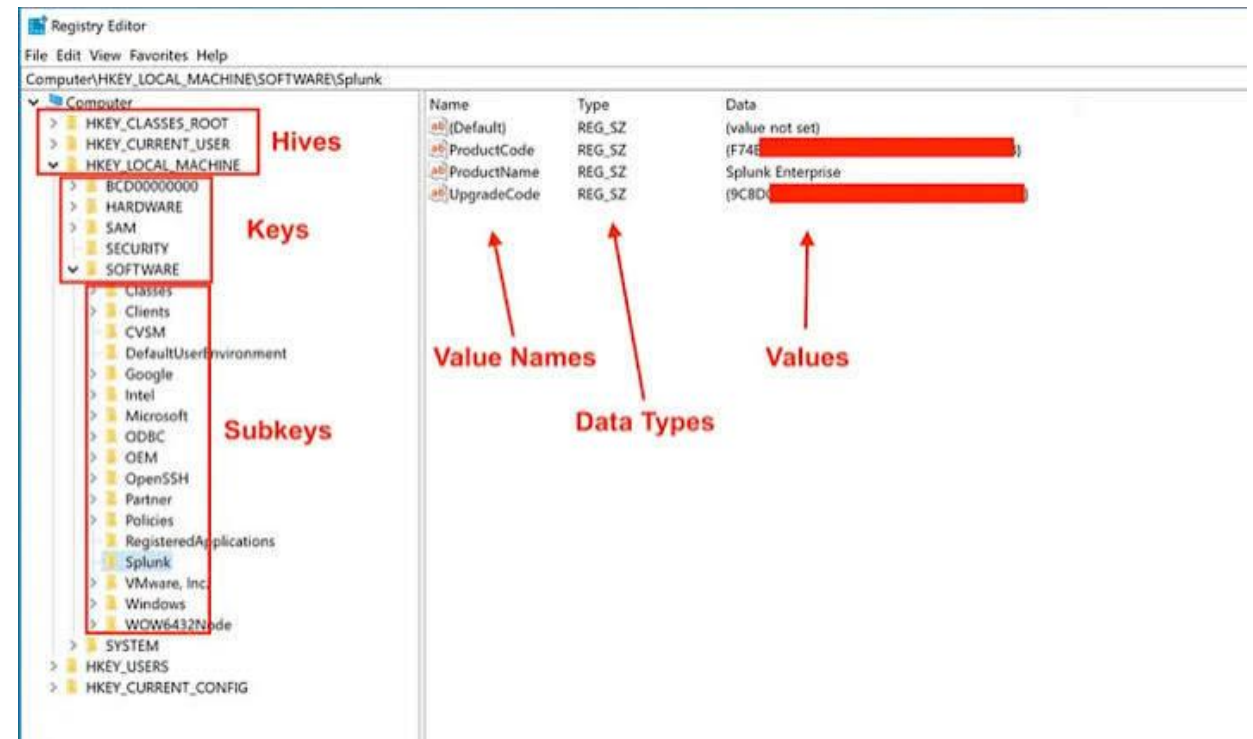
The **Windows Registry** is a hierarchical database that stores configuration and operational information used by Windows and installed applications.

Forensic investigators examine Registry data because it can preserve information about **users, applications, devices, system configuration and activity**.

Major Registry Hives

- NTUSER.DAT
- SOFTWARE
- SYSTEM
- SAM
- SECURITY

Each hive contains different categories of information that can contribute to a forensic investigation.



UserAssist and Program Execution

UserAssist is a Windows Registry artefact associated with information about programs accessed through the Windows graphical user interface.

The information can help investigators identify applications that a user interacted with.

Forensic Value

UserAssist information can contribute to determining:

- Which programs were accessed
- Which user account was involved
- Approximate activity timing
- Possible application usage

Some UserAssist values are stored using **ROT13 encoding**, which must be decoded during analysis.

Important Point

UserAssist should be correlated with other artefacts rather than treated as independent proof of program execution.



USBSTOR and USB Device History

USBSTOR is a Windows Registry location associated with information about USB storage devices that have been connected to a system.

USB device artefacts can provide information about previously connected storage devices.

Potential Information

- Device name
- Device identification
- Serial number
- Connection-related information
- Device history



Forensic Importance

USB artefacts can help investigators determine whether a particular removable storage device was associated with a Windows system.

This information can be especially useful when investigating **data transfer or removable-media activity**.



RecentDocs and ShimCache

RecentDocs is a Windows Registry artefact that can contain information about files that were recently accessed or opened by users.

It can provide context about **recent document activity**.

ShimCache, also known as Application Compatibility Cache, contains information that can assist forensic investigators in analysing application-related activity.

It may provide evidence associated with programs that have existed or interacted with the system.

Forensic Principle

Registry artefacts should be interpreted together with other evidence such as Prefetch, event logs, file-system timestamps and user activity.



Run and RunOnce

Run and **RunOnce** Registry locations can contain entries that cause programs to execute automatically during specified Windows startup or user-logon conditions.

Forensic Importance

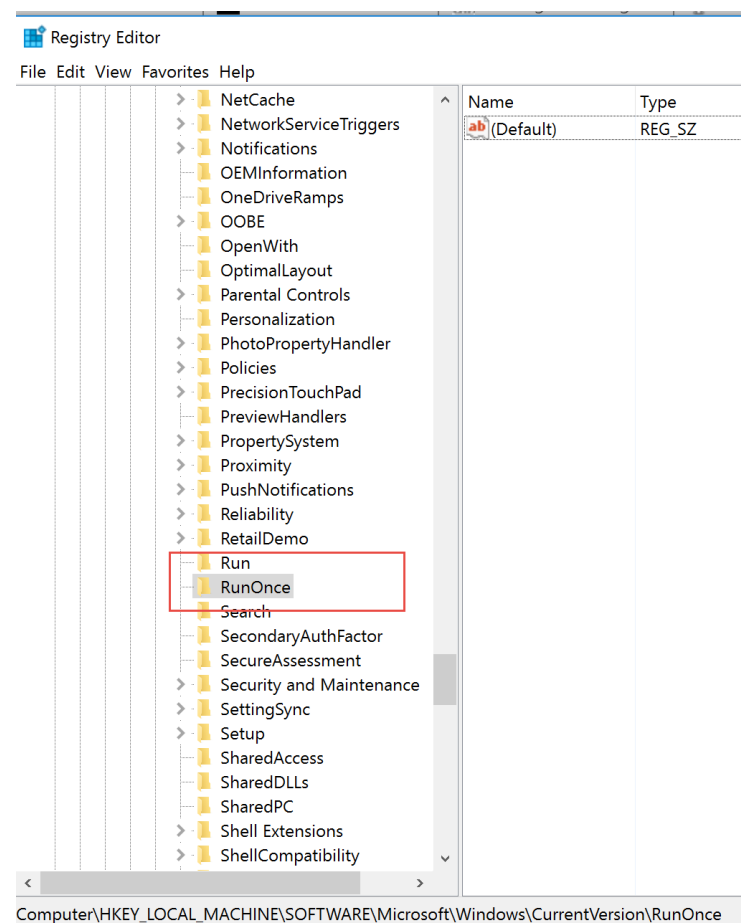
These locations are important when investigating **persistence mechanisms**.

An investigator may examine:

- Program paths
- Executable names
- User context
- Creation or modification information
- Whether an entry appears legitimate or suspicious

Investigation Objective

Determine whether an application has been configured to **automatically execute without normal user interaction**.



TypedURLs

TypedURLs is a Windows Registry artefact associated with URLs manually entered into supported Windows browser environments.

It can provide information about websites that were typed into the address bar.

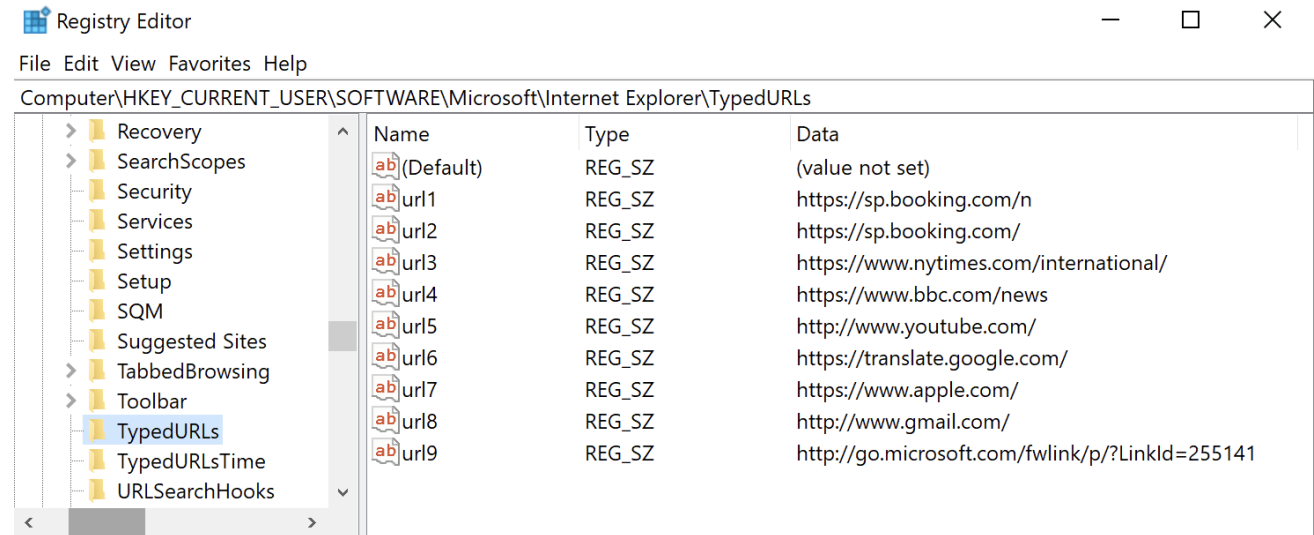
Forensic Value

This information can contribute to investigations involving:

- Web activity
- Suspicious websites
- Malicious downloads
- Phishing activity
- User browsing behaviour

Important Point

Browser history and other web artefacts should be examined alongside Registry information to establish a more complete browsing timeline.



Name	Type	Data
(Default)	REG_SZ	(value not set)
url1	REG_SZ	https://sp.booking.com/n
url2	REG_SZ	https://sp.booking.com/
url3	REG_SZ	https://www.nytimes.com/international/
url4	REG_SZ	https://www.bbc.com/news
url5	REG_SZ	http://www.youtube.com/
url6	REG_SZ	https://translate.google.com/
url7	REG_SZ	https://www.apple.com/
url8	REG_SZ	http://www.gmail.com/
url9	REG_SZ	http://go.microsoft.com/fwlink/p/?LinkId=255141



Windows Event Log Analysis



Windows Event Logs

Windows Event Logs record events generated by the operating system, applications and security components. These logs provide a chronological record of activities such as authentication, account changes, services and system events. The primary log categories include:

Security Log

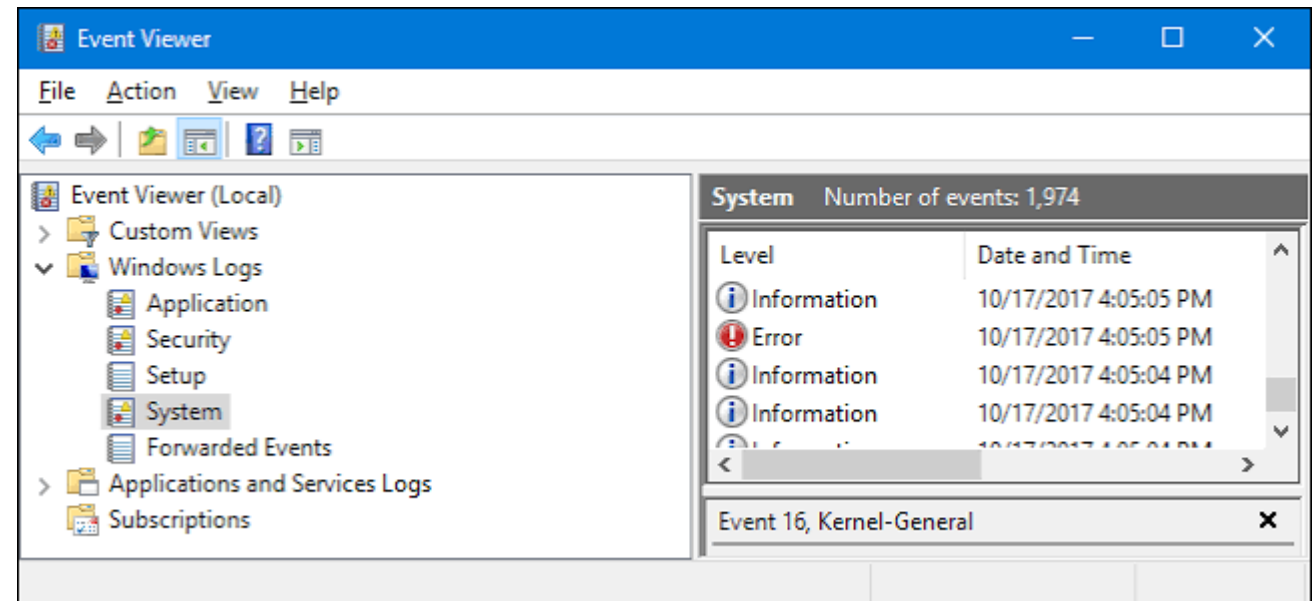
Authentication and security-related activity.

System Log

Operating-system and service-related activity.

Application Log

Events generated by applications.



Windows event logs are commonly stored using the **.evtx** format.



Security Event Logs

Security Events

The Windows Security log contains events related to authentication, account management and security activity.

Important Event IDs include:

- **4624** - Successful logon
- **4625** - Failed logon
- **4720** - User account created
- **4728 / 4732** - User added to security-related groups
- **4768 / 4769** - Kerberos authentication activity
- **4776** - NTLM authentication

Forensic Importance

- Security events can help reconstruct:
- Login attempts
- Successful authentication
- Account creation
- Privilege-related activity
- Authentication mechanisms



System and Application Logs

System Log

The System log records events associated with Windows operating-system components and services.

Event ID 7045 is associated with the installation of a new service.

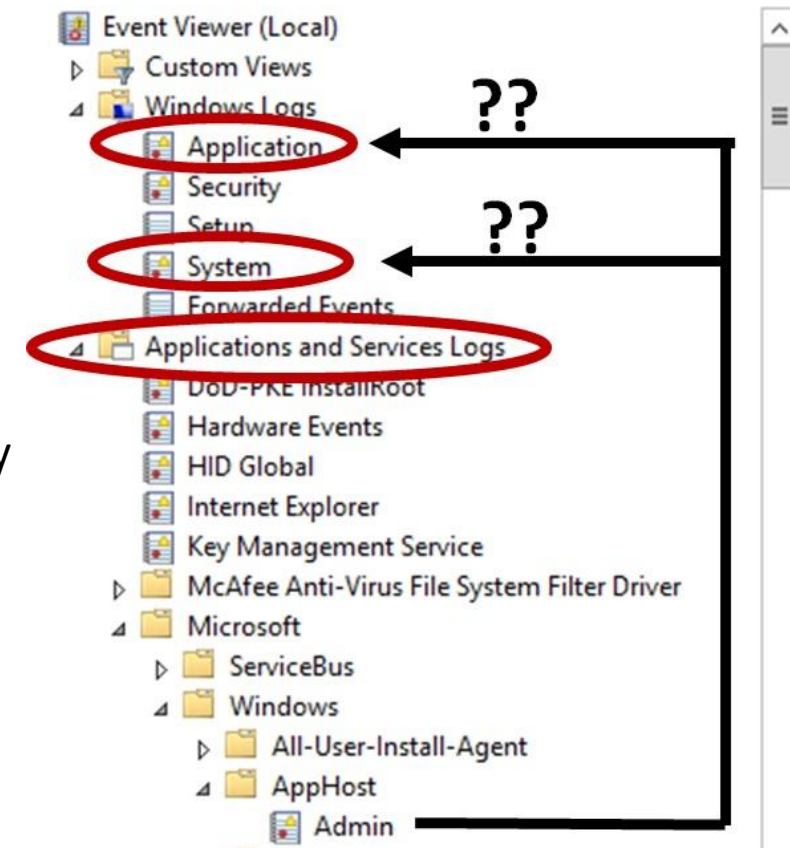
Application Log

The Application log contains events generated by installed applications and software components.

Forensic Importance

These logs can help investigators correlate application and system activity with other evidence.

For example, a newly installed service combined with unusual authentication activity may require further investigation.



Windows Operating System Artefacts



Prefetch Files

Prefetch is a Windows mechanism designed to improve application startup performance by recording information related to application execution.

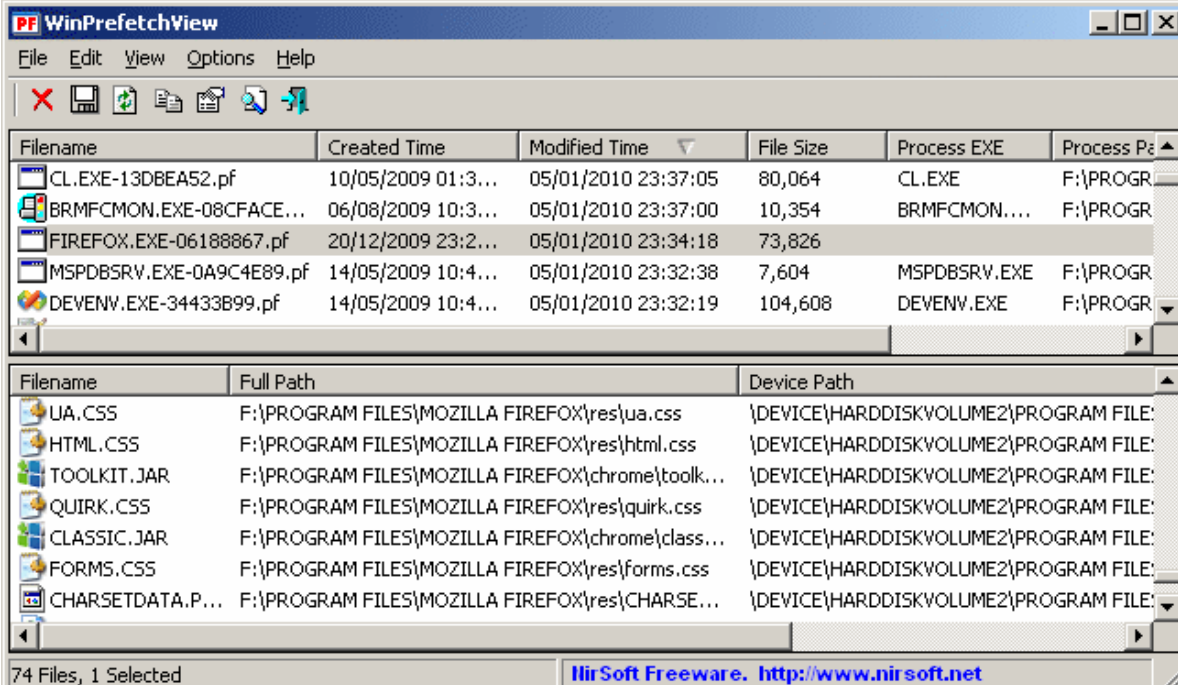
Prefetch files are generally stored with the **.pf** extension in:
C:\Windows\Prefetch

Forensic Value

Prefetch information can provide evidence associated with application execution, including applications that may no longer be present on the system.

Investigation Use

- Prefetch can be correlated with:
- File-system timestamps
- User activity
- Registry artefacts
- Event logs



WinPrefetchView

Filename	Created Time	Modified Time	File Size	Process EXE	Process Pa
CL.EXE-13DBEA52.pf	10/05/2009 01:3...	05/01/2010 23:37:05	80,064	CL.EXE	F:\PROGR
BRMFCMON.EXE-08CFACE...	06/08/2009 10:3...	05/01/2010 23:37:00	10,354	BRMFCMON...	F:\PROGR
FIREFOX.EXE-06188867.pf	20/12/2009 23:2...	05/01/2010 23:34:18	73,826		
MSPDBSRV.EXE-0A9C4E89.pf	14/05/2009 10:4...	05/01/2010 23:32:38	7,604	MSPDBSRV.EXE	F:\PROGR
DEVENV.EXE-34433B99.pf	14/05/2009 10:4...	05/01/2010 23:32:19	104,608	DEVENV.EXE	F:\PROGR

Filename	Full Path	Device Path
UA.CSS	F:\PROGRAM FILES\MOZILLA FIREFOX\res\ua.css	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:
HTML.CSS	F:\PROGRAM FILES\MOZILLA FIREFOX\res\html.css	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:
TOOLKIT.JAR	F:\PROGRAM FILES\MOZILLA FIREFOX\chrome\toolk...	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:
QUIRK.CSS	F:\PROGRAM FILES\MOZILLA FIREFOX\res\quirk.css	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:
CLASSIC.JAR	F:\PROGRAM FILES\MOZILLA FIREFOX\chrome\class...	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:
FORMS.CSS	F:\PROGRAM FILES\MOZILLA FIREFOX\res\forms.css	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:
CHARSETDATA.P...	F:\PROGRAM FILES\MOZILLA FIREFOX\res\CHARSE...	\DEVICE\HARDDISKVOLUME2\PROGRAM FILE:

74 Files, 1 Selected | NirSoft Freeware. <http://www.nirsoft.net>



LNK Files and Shellbags

LNK files are Windows shortcut files that can contain information about recently accessed files and their source locations.

They may provide information about:

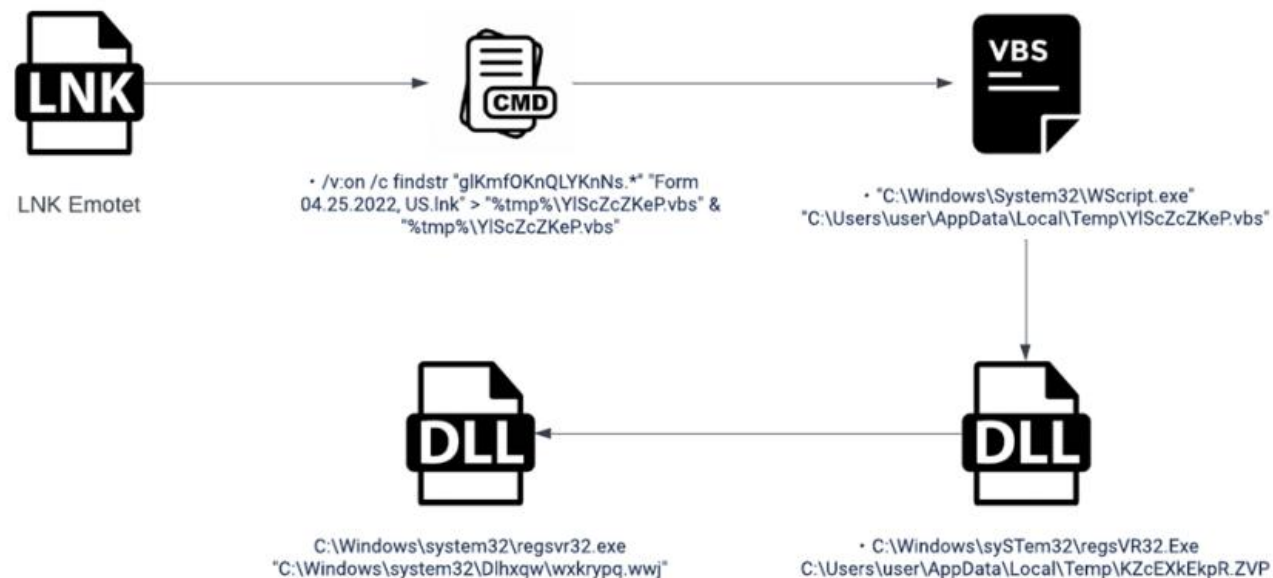
- File paths
- Target files
- User activity
- Access history

Shellbags are Windows artefacts associated with directory-browsing activity.

They can provide information about directories that users have interacted with, including directories that may no longer exist.

Forensic Value

Together, these artefacts can help reconstruct aspects of **user file and directory activity**.



Recycle Bin, Thumbcache and Jump Lists

Recycle Bin

Windows maintains information associated with deleted files in the Recycle Bin.
The relevant artefacts include:

\$R - File data

\$I - Metadata such as original path and deletion information



Thumbcache stores thumbnail representations generated for images and other media.
It may provide evidence of media that was viewed on a system.

Jump Lists contain information associated with recently accessed items and application activity.

Forensic Value

These artefacts can provide additional evidence about **files, media and application usage**.



Linux File System Forensics



Linux Forensics

Linux forensics is the examination of Linux file systems, logs, user information, command history, scheduled tasks and network artefacts to reconstruct system activity.

Linux stores forensic information across multiple locations rather than relying on a single central repository.

Major Areas

- EXT4 file system
- System and authentication logs
- Bash history
- User account files
- Cron jobs
- Login history
- Network information



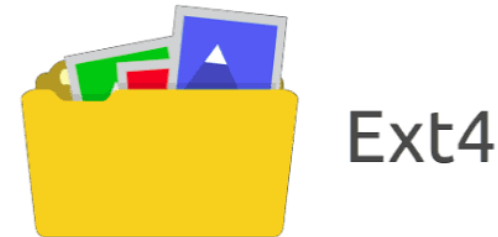
EXT4 File System and Inodes

EXT4 is a commonly used Linux file system.
A key structure within EXT4 is the **inode**.

Inode

An inode stores metadata associated with a file, such as:

- File type
- Permissions
- Ownership
- File size
- Timestamps
- References to data blocks



Forensic Importance

Inode information helps investigators understand **file metadata, ownership and file-system activity**.



Linux Log Files

Linux systems maintain logs that record system, authentication and application activity.

Important Log Sources

/var/log/auth.log

Authentication and privilege-related activity such as SSH and sudo.

/var/log/syslog

General system activity.

/var/log/kern.log

Kernel-related events.

/var/log/apache2/access.log

Web-server access activity.

Forensic Importance

Log analysis can help establish **when an event occurred, which account was involved and what activity took place.**

```
spark@workstation:/var/log$ ls
alternatives.log      containers            kern.log.1
alternatives.log.1    cups                kern.log.2.gz
alternatives.log.2.gz dist-upgrade         kern.log.3.gz
alternatives.log.3.gz dmesg               kern.log.4.gz
appport.log           dmesg.0            lastlog
appport.log.1         dmesg.1.gz         openvpn
appport.log.2.gz      dmesg.2.gz         pods
appport.log.3.gz      dmesg.3.gz         private
appport.log.4.gz      dmesg.4.gz         speech-dispatcher
appport.log.5.gz      dpkg.log            syslog
appport.log.6.gz      dpkg.log.1          syslog.1
apt                   dpkg.log.2.gz       syslog.2.gz
auth.log              dpkg.log.3.gz       syslog.3.gz
auth.log.1            faillog             syslog.4.gz
auth.log.2.gz         fontconfig.log      teamviewer15
auth.log.3.gz         gdm3               ubuntu-advantage.log
auth.log.4.gz         gpu-manager.log     ubuntu-advantage-timer.log
bootstrap.log         hp                 ubuntu-advantage-timer.log.1
btmtp                 installer          unattended-upgrades
btmtp.1               journal            wtmp
clamav                kern.log
spark@workstation:/var/log$
```



Bash History

Bash history records commands entered by users through the Bash shell. It can provide valuable information about administrative and user activity performed through the command line.

Forensic Value

Command history may reveal:

- Commands executed
- Files accessed
- System configuration changes
- Network commands
- Administrative activity



Important Consideration

Attackers may attempt to delete or modify command history. Therefore, the absence of a command from history does not necessarily prove that the command was never executed.



Linux User and Password Information

/etc/passwd

The /etc/passwd file contains information about local user accounts, including usernames, user IDs and other account-related information.

/etc/shadow

The /etc/shadow file stores password-related information and is normally protected from ordinary users.

Forensic Importance

These files help investigators understand:

- User accounts
- Account identifiers
- Account configuration
- Authentication-related information

```
tecmin@TecMint - $ finger tecmint
Login: tecmint                Name: TecMint
Directory: /home/tecmin      Shell: /bin/bash
On since Fri Sep 22 10:39 (IST) on tty8 from :0
      2 hours 22 minutes idle
No mail.
No Plan.

tecmin@TecMint - $ w
12:59:52 up 2:23, 1 user, load average: 0.86, 0.56, 0.52
USER      TTY      FROM          LOGIN@      IDLE        JCPU   PCPU   WHAT
tecmin    tty8      :0             10:39       2:23m      5:05    0.48s  cinnamon-

tecmin@TecMint - $
```

→ Show Complete Account Info

→ Show User Login Info and Activity

Access to sensitive authentication data must be appropriately controlled during forensic examination.



Cron Jobs and Scheduled Tasks

Cron is a Linux scheduling mechanism used to execute commands or programs automatically at specified times or intervals.

Forensic Importance

Scheduled tasks can be legitimate administrative functions, but they can also be used as a **persistence mechanism**.

Investigators may examine:

- `/etc/cron*`
- User crontabs
- `crontab -l`
- Commands configured for execution
- Execution schedules



Investigation Objective

Determine whether scheduled tasks are expected system activities or potentially suspicious mechanisms.



Linux Login and Network Artefacts

Login History

Linux commands such as:

last — Displays historical login information.

lastb — Displays recorded failed login attempts.

These can help investigators analyse authentication activity.

Network Artefacts

/proc/net/tcp contains information associated with TCP network connections.

Forensic Value

Combining login and network information can help establish relationships between:

User activity → Authentication → Network activity



Windows vs Linux Forensics

Area	Windows	Linux
File System	NTFS	EXT4
Key Metadata	MFT	Inodes
Central Configuration	Registry	Configuration files
Authentication Logs	Event Logs	/var/log/auth.log
Command History	Application/user artefacts	Bash history
Scheduled Tasks	Task Scheduler / Registry artefacts	Cron
Login Analysis	Security Event Logs	last, lastb
Network Artefacts	Event/OS artefacts	/proc/net/tcp



Correlation and Timeline Analysis



Correlating Forensic Artefacts

A single artefact rarely provides a complete explanation of an incident.
Investigators should correlate information from multiple sources to establish a reliable sequence of events.

Example Investigation Flow

Event Log → Successful login

Registry → USB device identified

MFT → File accessed

LNK / RecentDocs → Document activity

Prefetch → Application execution

Timeline → Events arranged chronologically

Objective

Multiple artefacts → **Correlated evidence** → **Reconstructed activity**



Integrated Investigation Example

Scenario

A Windows workstation is suspected of being used to access sensitive information and transfer it to a removable device.

An investigator may examine:

Security Logs: To identify successful and failed logins.

USBSTOR: To identify connected USB devices.

MFT and MACB timestamps: To examine file activity.

RecentDocs / LNK files: To identify document access.

Prefetch: To examine application execution.

Recycle Bin: To examine deleted files.

Investigation Goal

Correlating these artefacts can help reconstruct **who** accessed the system, **what** activity occurred and **when** the activity took place.



Thank you

